TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Colour Fastness Tests of Different Dyes

Washing, rubbing and light fastness graded on the ISO grey scales (1–5) and blue-wool scale (1–8) — the verdict sheet for every dyeing in this course.

TE-3106 · TEXTILE CHEMISTRY & COLORATION-II SESSIONAL · EXPERIMENT 5 · LEVEL 3 TERM 1

1. Aim and theory

Aim: to assess the colour fastness of dyed samples (from Experiments 2 and 3) to **washing, rubbing and light**, grading change of shade and staining against the ISO grey scales. Fastness is graded **1 (very poor) to 5 (excellent)** in half-steps (e.g. 3–4). Two grey scales exist: **change of shade** (contrast between treated and original sample) and **staining** (contrast between stained and unstained adjacent multifibre fabric). Grading is done under D65 light in a viewing cabinet, by eye against the scale pairs — or instrumentally as ΔE converted to a grey-scale rating.

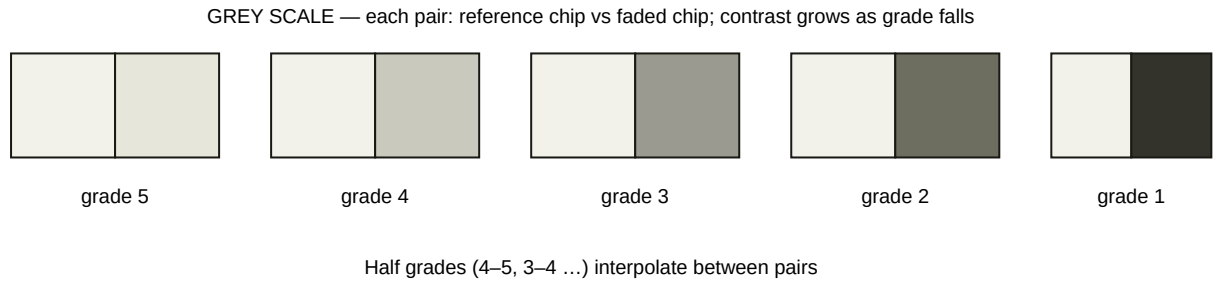


Fig. 1 — Principle of the grey scale for change of shade (staining scale works identically on white multifibre).

2. Test A — wash fastness (ISO 105-C06 principle)

1. Specimen 10 × 4 cm sewn with multifibre adjacent fabric (acetate, cotton, nylon, polyester, acrylic, wool stripes).
2. Bath: standard soap/detergent 4 g/L, LR 50:1, steel balls for mechanical action, 40 °C (A1S) or 60 °C (C1S) for 30 min in a wash-fastness tester.
3. Rinse, dry in air ≤ 60 °C. Grade change of shade of the specimen and staining of each multifibre stripe.

3. Test B — rub fastness (ISO 105-X12 principle)

A crockmeter rubs a standard white cotton cloth against the specimen under 9 N load for 10 cycles over 10 cm, dry and then wet (cloth at ~100% pick-up). Grade the **staining** of the rubbing cloth. Wet rub is almost always worse: water swells the fibre and lubricates pigment/dye transfer — the pigment-printed samples of Topic 10 show the largest dry–wet gap.

4. Test C — light fastness (ISO 105-B02 principle)

Specimen and blue-wool references 1–8 are exposed together to a xenon-arc (or daylight) source with part of each masked. Exposure continues until the specimen's exposed/unexposed contrast equals a set grey-scale step; the light fastness is the number of the blue-wool reference that faded to the same extent. Note: light fastness uses the **1–8 blue-wool scale**, not the 1–5 grey grades — the one scale difference students must state precisely.

5. Results (as recorded, samples from Experiments 2–3)

SAMPLE	WASH: CHANGE	WASH: STAINING (COTTON)	RUB DRY	RUB WET
Direct on cotton	3	2–3	4	3
Vat on cotton	4–5	4–5	4	3–4
Acid on wool	4	4 (wool stripe)	4–5	4
Chrome mordant, wool	4–5	4–5	4–5	4
Pigment on P/C	4–5	4–5	3	2–3

The table teaches the course in one view: covalent/entrapped systems (vat) and coordination lakes (chrome) top the wash column; the direct dye's weak forces show; and the pigment sample — perfect in wash — is exposed by rubbing, exactly as its surface-film mechanism predicts.

6. Precautions

- Grade in the D65 cabinet after conditioning, resting the eyes between samples; never grade under tube light at the bench.
- Multifibre must touch the specimen face fully; loose sewing gives false low staining.
- Report every stripe's staining, not only the worst — buyers specify per-fibre limits.

7. Practice (viva preparation)

1. Distinguish the two grey scales and the blue-wool scale: what does each measure, and what are their ranges?
2. Why is wet rub fastness lower than dry for the same sample?
3. A sample grades wash change 4–5 but staining 2. What is happening in the wash bath, and which after-treatment targets it?
4. State the buyer meaning of "4" as a minimum wash grade for shirting.

Answer key

1. Grey scale for change: fading/alteration of the specimen itself, 1–5. Grey scale for staining: colour transferred to adjacent undyed fabric, 1–5. Blue-wool scale: light fastness only, 1–8 against standard wool references.
2. Water swells cellulosic fibres, softens binder films and reduces friction resistance, so colorant particles and loosely held dye transfer more readily to the rubbing cloth.
3. Dye is leaving the specimen (little visible change because depth is high) and redepositing on adjacent fibres — unfixed/hydrolysed dye not removed in production soaping. Target: thorough hot soaping/wash-off and, for direct dyes, cationic fixing after-treatment.
4. Grade 4 = slight change/staining permitted — the common commercial pass level for apparel washing; below it, garments visibly fade or stain other laundry within a few washes.
